Complex Numbers

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Complex numbers

Definition 0.1 (Imaginary unit)**.** The **imaginary unit** i is a number whose square is -1:

$$i^2 \stackrel{\text{def}}{=} -1.$$

Apart from that one rule, i obeys the usual rules of arithmetic:

- sums and products with i are commutative;
- sums and products with i are associative;
- multiplication distributes over addition.

Axler (2024, Definition 1.1, p. 2) makes this idea rigorous: it defines a complex number as an ordered pair (a, b) of real numbers, written $a + bi$, defines addition and multiplication of such pairs, writes $0 + 1i$ as i, and leaves it to the reader to verify that $i^2 = -1$.

Example 0.1 (Powers of the imaginary unit)**.** The powers of i repeat in a cycle of four:

$$\begin{aligned} i^3 &= i^2 \cdot i && \text{(split off one factor of } i\text{)} \\ &= (-1) \cdot i && \text{(Definition 0.1)} \\ &= -i && \text{(multiply)} \end{aligned}$$

and

$$\begin{aligned} i^4 &= i^2 \cdot i^2 && \text{(split the power into two squares)} \\ &= (-1) \cdot (-1) && \text{(Definition 0.1, twice)} \\ &= 1 && \text{(multiply)} \end{aligned}$$

so the cycle starts again:

$$\begin{aligned} i^5 &= i^4 \cdot i && \text{(split off one factor of } i\text{)} \\ &= 1 \cdot i && (i^4 = 1) \\ &= i && \text{(multiply)} \end{aligned}$$

and the powers run $i, -1, -i, 1, i, \ldots$.

Example 0.2 (No real number squares to -1)**.** The imaginary unit is not a real number, because the square of every real number x is at least 0:

- if $x \geq 0$, then $x^2 = x \cdot x$ is a product of two nonnegative numbers, so $x^2 \geq 0$;
- if $x < 0$, then $x^2 = x \cdot x$ is a product of two negative numbers, so $x^2 > 0$.

For instance, $3^2 = 9$ and $(-3)^2 = 9$; neither is -1. So the equation $x^2 = -1$ has no real solution, and Definition 0.1 introduces a new number to solve it.

Theorem 0.1 (The number $-i$ also squares to -1)**.**

$$(-i)^2 = -1.$$

Proof

Proof.

$$\begin{aligned} (-i)^2 &= (-1)^2\, i^2 && (-i = (-1) \cdot i, \text{ and } (ab)^2 = a^2 b^2 \text{ because products commute and associate}) \\ &= 1 \cdot (-1) && ((-1)^2 = 1 \text{ and Definition 0.1}) \\ &= -1 && \text{(multiply)} \end{aligned}$$

□

Example 0.3 (Squaring i and $-i$ in R)**.** R has the imaginary unit as `1i`. The code below squares it and its negative.

```
squares <- c(i = (1i)^2, minus_i = (-1i)^2)
squares
#>       i minus_i
#> -1+0i -1+0i
```

Both squares have real part -1 and imaginary part 0, as Definition 0.1 and Theorem 0.1 say.

ℹ Remark (The rule does not tell i from $-i$)

Remark 0.1 (The rule does not tell i from $-i$). Because $-i$ obeys the same rule as i (Theorem 0.1), the rule $i^2 = -1$ does not single one of them out. The symbol i names one of these two square roots of -1, chosen once and for all.

Definition 0.2 (Complex number)**.** A **complex number** is a number of the form

$$z = a + b\,i,$$

where a and b are real numbers and i is the imaginary unit (Definition 0.1).

- The **real part** of z is $\operatorname{Re} z \stackrel{\text{def}}{=} a$.
- The **imaginary part** of z is $\operatorname{Im} z \stackrel{\text{def}}{=} b$.
- The set of all complex numbers is $\mathbb{C} \stackrel{\text{def}}{=} \{a + b\,i : a, b \in \mathbb{R}\}$.

Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal.

Axler (2024, Definition 1.1, p. 2) defines $\mathbb{C}$ as the set of ordered pairs (a, b) of real numbers, written $a + bi$, and Axler (2024, Definition 4.1, p. 120) defines the real and imaginary parts.

Example 0.4 (Real and imaginary parts)**.** For $z = 2 - 5\,i$, the real part is $\operatorname{Re} z = 2$ and the imaginary part is $\operatorname{Im} z = -5$: the imaginary part is the real number -5, not $-5\,i$.
The numbers $1 + i$ and $1 - i$ have the same real part, 1, but different imaginary parts, 1 and -1, so they are different complex numbers.
The real number 7 is the complex number $7 + 0\,i$, with real part 7 and imaginary part 0.

Theorem 0.2 (Every real number is a complex number)**.** *A complex number z is a real number exactly when $\operatorname{Im} z = 0$. In particular, every real number a is the complex number $a + 0\,i$, with $\operatorname{Re} a = a$ and $\operatorname{Im} a = 0$, so $\mathbb{R}$ is a subset of $\mathbb{C}$.*

ℹ Proof

Proof. Let a be a real number. Since $0\,i = 0$,

$$\begin{aligned} a + 0\,i &= a + 0 && (0\,i = 0) \\ &= a && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

so a is the complex number $a + 0\,i$, whose imaginary part is 0. The real part of $a + 0\,i$ is a, and by the equality rule in Definition 0.2 no other pair of real and imaginary parts

gives the same number.
Conversely, if $z = a + b\,i$ has $b = \operatorname{Im} z = 0$, then $z = a + 0\,i = a$ by the same calculation, so z is real. □

Example 0.5 (Which complex numbers are real)**.** R stores complex numbers as a real part and an imaginary part. The code below converts the real number 7 to a complex number and reads the parts of three complex numbers.

```
z <- c(as.complex(7), 2 - 5i, 3 + 0i)
data.frame(z = z, re = Re(z), im = Im(z), is_real = Im(z) == 0)
#>      z re im is_real
#> 1 7+0i  7  0    TRUE
#> 2 2-5i  2 -5   FALSE
#> 3 3+0i  3  0    TRUE
```

The real number 7 becomes the complex number 7+0i, with imaginary part 0. Of the 3 numbers, 2 have imaginary part 0, and by Theorem 0.2 these are the real ones. The number $2 - 5\,i$ is not real, because its imaginary part is -5.

Theorem 0.3 (Adding and multiplying complex numbers)**.** *For real numbers a, b, c, d:*

$$(a + b\,i) + (c + d\,i) = (a + c) + (b + d)\,i$$

and

$$(a + b\,i)(c + d\,i) = (ac - bd) + (ad + bc)\,i.$$

Proof

Proof. **Sum.**

$$\begin{aligned}(a + b\,i) + (c + d\,i) &= a + c + b\,i + d\,i && \text{(rearrange the terms)}\\ &= (a + c) + (b + d)\,i && \text{(factor out } i\text{)}\end{aligned}$$

Product.

$$\begin{aligned}(a + b\,i)(c + d\,i) &= ac + ad\,i + bc\,i + bd\,i^2 && \text{(distribute)}\\ &= ac + ad\,i + bc\,i - bd && \text{(Definition 0.1)}\\ &= (ac - bd) + (ad + bc)\,i && \text{(group the real terms and the terms with } i\text{)}\end{aligned}$$

□

Axler (2024, Definition 1.1, p. 2) takes these two formulas as the definitions of addition and multiplication.

Example 0.6 (Adding and multiplying two complex numbers)**.** Let $w = 1 + 2\,i$ and $z = 3 - i$. Their sum is

$$\begin{aligned} w + z &= (1 + 3) + (2 + (-1))\,i && \text{(Theorem 0.3, sum)} \\ &= 4 + i && \text{(add)} \end{aligned}$$

and their product, multiplying out directly, is

$$\begin{aligned} wz &= 1 \cdot 3 + 1 \cdot (-i) + 2\,i \cdot 3 + 2\,i \cdot (-i) && \text{(distribute)} \\ &= 3 - i + 6\,i - 2\,i^2 && \text{(multiply)} \\ &= 3 - i + 6\,i + 2 && \text{(Definition 0.1)} \\ &= (3 + 2) + (-1 + 6)\,i && \text{(group the real terms and the terms with } i\text{)} \\ &= 5 + 5\,i && \text{(add)} \end{aligned}$$

The product formula in Theorem 0.3 gives the same answer: with $a = 1$, $b = 2$, $c = 3$ and $d = -1$,

$$\begin{aligned} ac - bd &= 3 - (-2) \\ &= 5 \end{aligned}$$

and

$$\begin{aligned} ad + bc &= -1 + 6 \\ &= 5. \end{aligned}$$

Definition 0.3 (Complex conjugate)**.** The **complex conjugate** of a complex number $z = a + b\,i$ (Definition 0.2) is

$$\overline{z} \stackrel{\text{def}}{=} a - b\,i.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition, as $\overline{z} = \operatorname{Re} z - (\operatorname{Im} z)\,i$.

Example 0.7 (Complex conjugates)**.**

- $\overline{3 + 4\,i} = 3 - 4\,i$.

- $$\begin{aligned}\overline{-2\,i} &= \overline{0 + (-2)\,i} \\ &= 0 - (-2)\,i \\ &= 2\,i.\end{aligned}$$

- $$\begin{aligned}\overline{5} &= \overline{5 + 0\,i} \\ &= 5 - 0\,i \\ &= 5\,:\end{aligned}$$

  a real number is its own complex conjugate. A complex number with a nonzero imaginary part, such as $3 + 4\,i$, is not.

Definition 0.4 (Absolute value (modulus) of a complex number)**.** The **absolute value**, or **modulus**, of a complex number $z = a + b\,i$ (Definition 0.2) is

$$|z| \stackrel{\text{def}}{=} \sqrt{a^2 + b^2}.$$

Axler (2024, Definition 4.2, p. 120) gives the same definition.

Example 0.8 (Absolute values of complex numbers)**.**

- $$\begin{aligned}|3 + 4\,i| &= \sqrt{3^2 + 4^2} \\ &= \sqrt{25} \\ &= 5.\end{aligned}$$

- $$\begin{aligned}|-2\,i| &= \sqrt{0^2 + (-2)^2} \\ &= \sqrt{4} \\ &= 2.\end{aligned}$$

- For a real number $a = a + 0\,i$,

  $$\begin{aligned}|a| &= \sqrt{a^2 + 0^2} \\ &= \sqrt{a^2},\end{aligned}$$

  which is the usual absolute value of a; for instance,

  $$\begin{aligned}|-3| &= \sqrt{9} \\ &= 3.\end{aligned}$$

Theorem 0.4 (A complex number times its conjugate)**.** *For every complex number z,*

$$z\,\overline{z} = |z|^2.$$

In particular, $z\,\overline{z}$ is a nonnegative real number.

Proof

Proof. Write $z = a + b\,i$ with a, b real. Then

$$\begin{aligned} z\,\overline{z} &= (a + b\,i)(a - b\,i) && \text{(Definition 0.3)} \\ &= a^2 - ab\,i + ab\,i - b^2\,i^2 && \text{(distribute)} \\ &= a^2 - b^2\,i^2 && \text{(cancel } ab\,i\text{)} \\ &= a^2 + b^2 && \text{(Definition 0.1)} \\ &= |z|^2 && \text{(Definition 0.4)} \end{aligned}$$

□

Axler (2024, result 4.4, p. 121) lists this identity among the properties of complex numbers.

Example 0.9 (Multiplying $3 + 4\,i$ by its conjugate)**.**

$$\begin{aligned} &(3 + 4\,i)(3 - 4\,i) \\ &= 9 - 12\,i + 12\,i - 16\,i^2 && \text{(distribute)} \\ &= 9 - 16\,i^2 && \text{(cancel } 12\,i\text{)} \\ &= 9 + 16 && \text{(Definition 0.1)} \\ &= 25 && \text{(add)} \end{aligned}$$

which is $|3 + 4\,i|^2 = 5^2$ from Example 0.8, as Theorem 0.4 says.

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